

WENTWORTH LIU

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Data Scientist · Scientific Data Workflows · Data Quality & Automation · Statistical Modelling

SUMMARY

Data Scientist with 2+ years of experience building reliable data workflows, automated checks, and analytical tools for complex scientific and operational datasets. Combines a Computational Materials Science research background with experience in multi-source data integration, data-quality monitoring, configurable feature systems, and decision support at Li Auto and ByteDance. Strong in Python and SQL, with a careful, structured approach to data definitions, validation, workflow standardisation, and technical communication.

EXPERIENCE

Li Auto

Data Scientist, Battery Intelligence Department

Beijing, China

Jul 2024 – Present

- **Scientific Data Quality & Standardisation.** Built a configurable feature system for multi-dimensional fleet telemetry using custom event definitions, multi-operator aggregations, and cross-platform normalisation; deployed reusable workflows to an internal ML platform and improved feature-iteration efficiency by ~85%.
- **Risk Data Analysis & QA.** Built an explainable risk-ranking pipeline for thermal-runaway detection across multiple battery platforms, generating vehicle-level risk scores and feature-level drivers; achieved 93% Recall@2% on retrospectively validated high-risk cases while limiting review to the highest-risk 2% of vehicles.
- **Analytical Workflow Development.** Used quasi-causal analysis on million-scale fleet telemetry to identify high-risk SOH-degradation cohorts and actionable drivers; productised intervention analysis across multiple vehicle models to support consistent assessment and targeted decision-making.
- **Data Classification & Review Automation.** Built a hybrid rule-based and statistical classifier for battery-quality work orders across 30+ fault categories, combining text matching, TF-IDF, and Random Forest; achieved Weighted F1 > 0.90 and used confidence-based routing to focus review effort on ambiguous cases.

ByteDance

Data Analytics Intern, Douyin Market Strategy

Beijing, China

May – Sep 2023

- **Multi-source Data Integration & Monitoring.** Integrated consumption, dissemination, and search signals across multiple data platforms into an IP popularity measurement system; added automated pipeline monitoring to detect upstream data instability and support reliable recurring analysis.
- **Metric Definition & Analytical Reporting.** Redesigned performance metrics and built a multi-dimensional content taxonomy for Qishui Music; used funnel segmentation and KOL audience-saturation modelling to diagnose conversion decline and inform content and marketing-resource decisions.

Dongwu Securities

Equity Research Intern, Mechanical Sector

China

Dec 2021 – Mar 2022

- **Data-led Research & Reporting.** Combined company fundamentals, competitive analysis, and market dynamics with stock-price modelling to deliver structured investment recommendations for Harmonic Drive Systems.

EDUCATION

Nanjing University

MSc Computational Materials Science

Sep 2021 – Jun 2024

- Developed Skeaf, a 3D Fermi-surface visualisation tool combining HDBSCAN clustering with quantum-oscillation data; first author, Physical Review B (2024); co-author, Physical Review B (2024) and Physical Review Materials (2023); Patent: CN115795266A.

Nanjing University

BSc Materials Science & Engineering (Outstanding Graduate) · GPA: 4.6/5.0 · Top 1% of cohort

Sep 2017 – Jul 2021

- National Scholarship (top 0.15%) · MCM/ICM 2020 Honorable Mention (Team Captain) · Sakura Science Exchange Programme, Tohoku University, Japan (5 selected university-wide).

SKILLS

Programming

Python · SQL/HiveSQL · C++ · MATLAB · Git

Data Quality & Workflow

Data Validation · QA/QC Principles · Multi-source Data Integration · Data Standardisation · Automated Monitoring · Workflow Automation · Data Pipelines · Visualisation · Reporting

Analytics & Modelling

Statistical Modelling · Machine Learning · Predictive Modelling · Causal Inference · Feature Engineering · NLP · Time Series · LightGBM · scikit-learn

Domain & Lang.

Scientific Computing · Materials Science · EV Analytics · English (TOEFL 104) · Mandarin/Cantonese (Native) · Japanese (JLPT N1)